Seat No. ____________

# SAVITRIBAI PHULE PUNE UNIVERSITY

## Deep Learning (410251)

## BE (Computer Engineering) --- Semester VIII --- 2019 Pattern

**Total Marks: 70 | Time: 2½ Hours**

**Instructions:**

1. Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
2. Figures to the right indicate full marks.
3. Neat diagrams must be drawn wherever necessary.
4. Make suitable assumptions whenever necessary.

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## Unit III --- Convolution Neural Network --- 18 Marks

**Q1) Solve any one of the following:**

**a) Draw and explain the architecture of Convolutional Neural Network (CNN) in detail. Label all layers and explain the role of each.** [6]

**b) Explain the working of a Convolution Layer. Discuss how convolution operations with different filters extract features. How do stride and padding affect the output dimensions?** [6]

**c) Explain Pooling Layers and their types. Compare Max Pooling, Average Pooling, and Global Pooling. When is each preferred?** [6]

OR

**Q2)** Solve **any one** of the following:

**a) Explain all the features of the Pooling Layer. Discuss spatial invariance, dimensionality reduction, and overfitting prevention.** [6]

**b) Explain Local Response Normalization (LRN) in CNNs. How does LRN help in generalization? Compare LRN with Batch Normalization.** [6]

**c) Explain the ReLU Layer in detail. Discuss its advantages over Sigmoid and Tanh. What is the dying ReLU problem and how does Leaky ReLU address it?** [6]

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## Unit IV --- Recurrent Neural Network --- 17 Marks

**Q3) Solve any one of the following:**

**a) Explain Recursive Neural Network. How is it different from Recurrent Neural Network? Give an application of each.** [6]

**b) Explain Long Short-Term Memory (LSTM) in detail. Draw and explain the cell architecture with forget, input, and output gates.** [6]

**c) Explain in brief the working of a Recurrent Neural Network. What is the role of the hidden state? How does an RNN handle variable-length sequences?** [5]

OR

**Q4)** Solve **any one** of the following:

**a) Differentiate between CNN and RNN. Compare their architectures, use cases, and training approaches.** [6]

**b) What are the challenges of long-term dependencies in RNNs? Explain the vanishing and exploding gradient problems. How do gated architectures (LSTM, GRU) alleviate these?** [6]

**c) Explain the Encoder-Decoder RNN model for sequence-to-sequence tasks. How is it used in machine translation? Discuss attention mechanism.** [5]

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## Unit V --- Deep Generative Models --- 18 Marks



**Q5) Solve any one of the following:**

**a) Explain Deep Generative Models. How do they differ from discriminative models? Discuss applications in image generation and data augmentation.** [6]

**b) Explain Boltzmann Machine in detail. Discuss its architecture, energy function, and training. How is a Restricted Boltzmann Machine (RBM) different?** [6]

**c) Explain Generative Adversarial Network (GAN) with an example. Describe the minimax objective function and the training process.** [6]

OR

**Q6)** Solve **any one** of the following:

**a) Explain Deep Belief Networks (DBN) in detail. How are they constructed from stacked RBMs? Discuss the greedy layer-wise pretraining approach.** [6]

**b) What is a Generative Adversarial Network? Explain its generator and discriminator components. Draw the architecture showing the adversarial training loop.** [6]

**c) Explain types of GAN: DCGAN, Conditional GAN (cGAN), and CycleGAN. Compare their architectures and applications.** [6]

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## Unit VI --- Reinforcement Learning --- 17 Marks

**Q7) Solve any one of the following:**

**a) Explain Markov Decision Process (MDP). Define the key components: states, actions, transition probabilities, reward function, and discount factor. How does policy iteration and value iteration work?** [6]

**b) Explain Deep Reinforcement Learning. How does combining deep neural networks with RL address the limitations of traditional RL? Discuss the Deep Q-Network (DQN) architecture.** [6]

**c) What are the challenges of Reinforcement Learning? Discuss the exploration-exploitation tradeoff, delayed rewards, and sample efficiency.** [5]

OR

**Q8)** Solve **any one** of the following:

**a) Explain the process of Deep Q-Learning. Describe the Q-learning update rule, experience replay, and target network. How do these stabilize training?** [6]

**b) Explain Reinforcement Learning for the Tic-Tac-Toe game. Model it as an MDP and describe the state representation, reward structure, and learning process.** [6]

**c) Explain the Dynamic Programming algorithm for Reinforcement Learning. Compare policy evaluation, policy improvement, and policy iteration.** [5]

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*** End of Paper ***